

Note 3 SVM.

1. Margins

intuition: find θ s.t. $\left\{ \begin{array}{l} \theta^T x^{(i)} \gg 0, \text{ whenever } y^{(i)} = 1 \\ \theta^T x^{(i)} \ll 0, \text{ whenever } y^{(i)} = 0 \end{array} \right.$

2. Notation

$$h_{w,b}(\vec{x}) = g(\vec{w}^T \vec{x} + b)$$

$$g(z) = 1 \text{ if } z \geq 0, \quad g(z) = -1, \text{ else}$$

3. Functional & geometric margin

$$\hat{y}^{(i)} = y^{(i)} (\vec{w}^T \vec{x} + b)$$

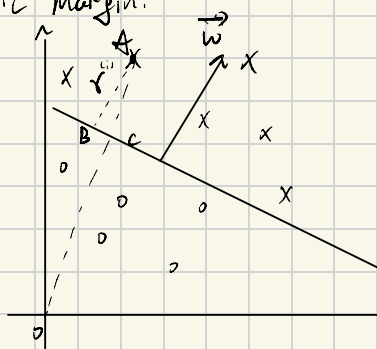
Same if $\|\vec{w}\| = 1$

property: $g(\vec{w}^T \vec{x} + b) = g(2\vec{w}^T \vec{x} + 2b)$

⇒ impose some sort of normalization

Functional margin $\hat{\gamma} = \min_i \hat{\gamma}^{(i)}$ → ensure all samples are categorized right.
 ↑
 try to maximize this minimum value

Geometric margin.



$B: x^{(i)} - \gamma^{(i)} \vec{w} / \|\vec{w}\|$
 on the decision boundary

$$\vec{w}^T (x^{(i)} - \gamma^{(i)} \vec{w} / \|\vec{w}\|) + b = 0$$

$$\Rightarrow \gamma^{(i)} = \left(\frac{\vec{w}}{\|\vec{w}\|} \right)^T x^{(i)} + \frac{b}{\|\vec{w}\|}$$

$$\Rightarrow \text{def } \gamma^{(i)} := \hat{\gamma}^{(i)} / y^{(i)}$$

$$\gamma = \min_i \gamma^{(i)}$$

⚡ Notation:
 $\hat{\gamma}$ geometric margin
 γ functional margin

4. The optimal margin classifier
A linearly separable training set:

$$\max_{\gamma, w, b} \gamma$$

$$\text{s.t. } y^{(i)}(w^T x^{(i)} + b) \geq \gamma \quad i=1, \dots, m$$

$$\|w\| = 1$$

functional margin = geometric margin.

$\Downarrow$ $\|w\|=1$ non-convex.

$$\max_{\gamma, w, b} \frac{\hat{\gamma}}{\|w\|}$$

$$\text{s.t. } y^{(i)}(w^T x^{(i)} + b) \geq \gamma \quad i=1, \dots, m$$

$\Downarrow$ rescale $w, b \Rightarrow$ set $\gamma=1$
 $\Downarrow \max \frac{1}{\|w\|} \Leftrightarrow \min \|w\|$

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

$$\text{s.t. } y^{(i)}(w^T x^{(i)} + b) \geq 1 \quad i=1, \dots, m$$

5. Lagrange duality.

primal optimization problem:

$$\min_w f(w)$$

$$\text{s.t. } g_i(w) \leq 0 \quad i=1, \dots, k$$

$$h_i(w) = 0 \quad i=1, \dots, l$$

def: generalized Lagrangian:

$$L(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$$

$$\theta_P(w) = \max_{\alpha, \beta, \alpha_i \geq 0} L(w, \alpha, \beta) \quad \text{given } w, \text{ max wrt } \alpha, \beta$$

$$\theta_P(w) = \begin{cases} f(w) & \text{if } w \text{ satisfies primal constraints} \\ \infty & \text{otherwise} \end{cases}$$

Consider: $\min_w \mathcal{Q}_0(w) = \min_w \max_{\alpha, \beta: \alpha_i \geq 0} L(w, \alpha, \beta)$ — Same as original primal prob

def. $p^* = \min_w \mathcal{Q}_0(w)$ — the value of the primal prob.

A slightly different prob:

$$\mathcal{Q}_0(\alpha, \beta) = \min_w L(w, \alpha, \beta) \quad \text{minimize wrt } w$$

D stands for "dual"

dual optimization prob.

$$\max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{Q}_0(\alpha, \beta) = \max_{\alpha, \beta: \alpha_i \geq 0} \min_w L(w, \alpha, \beta)$$

$$d^* = \max_{\alpha, \beta: \alpha_i \geq 0} \mathcal{Q}_0(w)$$

$$\text{Relation: } d^* = \max_{\alpha, \beta: \alpha_i \geq 0} \min_w L(w, \alpha, \beta) \leq \min_w \max_{\alpha, \beta: \alpha_i \geq 0} L(w, \alpha, \beta) = p^*$$

KKT Conditions: f & g 's convex

h 's affine

exists some w so that $g_i(w) < 0$ for all i

$\Rightarrow$ D must exist w^*, α^*, β^*

st. w^* is the solution to the primal problem

α^*, β^* are the solution to the dual problem.

$$\textcircled{1} p^* = d^* = L(w^*, \alpha^*, \beta^*)$$

$\textcircled{2} w^*, \alpha^*, \beta^*$ satisfy KKT conditions.

$$(i) \frac{\partial}{\partial w_i} L(w^*, \alpha^*, \beta^*) = 0 \quad i = 1, \dots, n$$

$$(ii) \frac{\partial}{\partial \beta_i} L(w^*, \alpha^*, \beta^*) = 0 \quad i = 1, \dots, d$$

$$(iii) \alpha_i^* g_i(w^*) = 0 \quad i = 1, \dots, k$$

$$(iv) g_i(w^*) \leq 0 \quad i = 1, \dots, k$$

$$(v) \alpha_i^* \geq 0 \quad i = 1, \dots, k$$

KKT dual complementarity condition: if $\alpha_i^* > 0$ then $g_i(w^*) = 0$

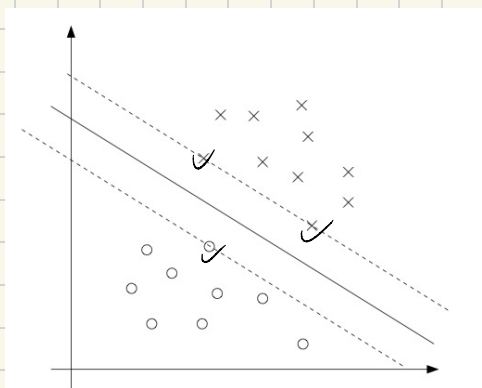
6. Optimal margin classifiers

Previously: $\min_{w, b} \frac{1}{2} \|w\|^2$

$$\text{s.t. } y^{(i)} (w^T x^{(i)} + b) \geq 1$$

Rewrite the constraints as: $f_i(w) = -y^{(i)} (w^T x^{(i)} + b) + 1 \leq 0$

have $\alpha_i > 0$ only for the training examples that have functional margin exactly equal to one



Only three of the α_i 's will be non-zero at the optimal solution to our optimization problem.

These three points are called support vectors.

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y^{(i)} (w^T x^{(i)} + b) - 1] \quad \dots (*)$$

Try to find the dual form of the problem: minimize L w.r.t w & b (for fixed α)

$$\nabla_w L(w, b, \alpha) = w - \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} = 0$$

$$w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$$

$$\frac{\partial}{\partial b} L(w, b, \alpha) = \sum_{i=1}^m \alpha_i y^{(i)} = 0$$

Plug $w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)}$ into (*).

$$L(w, b, \alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j (x^{(i)})^T x^{(j)} - \underbrace{b \sum_{i=1}^m \alpha_i y^{(i)}}_0$$

⇒ dual optimization problem.

$$\max_{\alpha} W(\alpha) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

$$\text{s.t. } \alpha_i \geq 0 \quad i=1, \dots, m$$

$$\sum_{i=1}^m \alpha_i y^{(i)} = 0$$

KKT conditions hold ⇒ solve the dual in lieu of solving the primal problem.

If we are indeed able to solve it:

$$\vec{\alpha}^* \checkmark \Rightarrow \vec{W}^* = \sum_i \alpha_i^* x^{(i)} y^{(i)}$$

$$\Rightarrow b^* = -\frac{1}{2} \left(\max_{i: y^{(i)}=1} W^{*T} x^{(i)} + \min_{i: y^{(i)}=-1} W^{*T} x^{(i)} \right)$$

After that, when we want to make a prediction at a new point input x

We would then calculate $w^T x + b$

$$\begin{aligned} W^T x + b &= \left(\sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \right)^T x + b \\ &= \sum_{i=1}^m \alpha_i y^{(i)} \langle x^{(i)}, x \rangle + b \end{aligned}$$

(Note: $x^{(i)}$ is vector-valued)

Since α_i 's will all be zero except for the support vectors, we need to find only the inner products between $\vec{x}$ and the support vectors.

7. Kernels

def. input attributes: $\vec{x}$ (not necessarily vector-valued,
↓ ϕ : feature mapping
input features $\phi(\vec{x})$)

Since the algorithm can be written entirely in terms of the inner product $\langle x, z \rangle$

$\Rightarrow$ we can replace all those inner products with $\langle \phi(x), \phi(z) \rangle$

$$K(x, z) = \phi(x)^T \phi(z)$$

i.e. $K(x, z) = (x^T z)^v$

rewrite as $K(x, z) = \left(\sum_{i=1}^n x_i z_i \right) \left(\sum_{j=1}^n x_j z_j \right)$

$$= \sum_{i=1}^n \sum_{j=1}^n x_i x_j z_i z_j$$

$$= \sum_{i,j=1}^n (x_i x_j) (z_i z_j)$$

$$\phi(x) = \begin{bmatrix} x_1 x_1 \\ x_1 x_2 \\ \vdots \\ x_2 x_1 \\ x_2 x_2 \\ \vdots \\ x_n x_1 \\ x_n x_2 \\ \vdots \\ x_n x_n \end{bmatrix}$$

Calculate $\phi(x)$ requires $O(n^2)$ time,
find $K(x, z)$ takes only $O(n)$ time!

More broadly, $K(x, z) = (x^T z + c)^d$
corresponds to a feature mapping to an $\binom{n+d}{d}$ feature space
Still, computing $K(x, z)$ takes only $O(n)$ time.

A slightly different view:

$$K(x, z) = \phi(x)^T \phi(z) \text{ measures how 'similar' are } \phi(x) \text{ and } \phi(z)$$

i.e. $K(x, z) = \exp\left(-\frac{\|x - z\|^v}{2\sigma^2}\right)$ — Gaussian kernel

(corresponds to an infinite dimensional feature mapping ϕ)

How can we tell if K is a valid kernel?

- Gain some insights:

Say, K is indeed a valid kernel (corresponding to some feature mapping ϕ).

Consider a finite set of m points $\{x^{(1)}, \dots, x^{(m)}\}$.

Let a square, m -by- m matrix K be defined
s.t.

$$K_{ij} = K(x^{(i)}, x^{(j)})$$

If K is a valid kernel.

$$K_{ij} = K(x^{(i)}, x^{(j)}) = \phi(x^{(i)})^T \phi(x^{(j)}) = \phi(x^{(j)})^T \phi(x^{(i)}) = K_{ji} \Rightarrow \text{Symmetric}$$

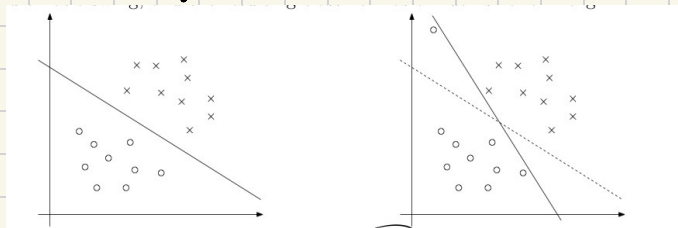
$$\begin{aligned} \forall \vec{z}: \quad \vec{z}^T K \vec{z} &= \sum_i \sum_j z_i K_{ij} z_j \\ &= \sum_i \sum_j z_i \phi(x^{(i)})^T \phi(x^{(j)}) z_j \\ &= \sum_i \sum_j z_i \sum_k \phi_k(x^{(i)}) \phi_k(x^{(j)}) z_j \\ &= \sum_k \sum_i \sum_j z_i \phi_k(x^{(i)}) \phi_k(x^{(j)}) z_j \\ &= \sum_k \left(\sum_i z_i \phi_k(x^{(i)}) \right)^2 \\ &\geq 0 \end{aligned}$$

$\Rightarrow K$ is positive semi-definite

Theorem (Mercer). Let $K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be given.

Then for K to be a valid (Mercer) kernel, it is necessary and sufficient that for any $\{x^{(1)} \dots x^{(m)}\}$, ($m < \infty$), the corresponding kernel matrix is symmetric positive semi-definite.

8. ~~Regularization~~ and the non-separable case
 Motivation: $\rightarrow$ high-dimensional: increased the likelihood of separable
 finding a separating hyper plane $\rightarrow$ susceptible to outliers



↳ regularization:

$$\min_{g, w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i$$

$$\text{s.t. } y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i \quad i=1, \dots, m$$

$$\xi_i \geq 0 \quad i=1, \dots, m$$

Note: C controls the relative weighting between two goals.

① $\|w\|^2$ small $\Rightarrow$ margin large

② most examples have functional margin at least 1

$$\text{Dual: } \max_{\alpha} W(x) = \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j=1}^m y^{(i)} y^{(j)} \alpha_i \alpha_j \langle x^{(i)}, x^{(j)} \rangle$$

$$\text{s.t. } 0 \leq \alpha_i \leq C \quad i=1, \dots, m$$

$$\sum_{i=1}^m \alpha_i y^{(i)} = 0$$

How to solve? $\rightarrow$ SMO

9. The SMO Algorithm.

Sequential Minimal Optimization.

9.1 Coordinate ascent

Consider solving an unconstrained optimization:

$$\max_{\alpha} W(\alpha_1, \alpha_2, \dots, \alpha_m)$$

$\Rightarrow$ Coordinate ascent:

Loop until convergence: {

For $i=1, \dots, m$, {

$$\alpha_i := \arg \max_{\alpha_i} W(\alpha_1, \dots, \alpha_{i-1}, \alpha_i, \alpha_{i+1}, \dots, \alpha_m)$$

}

}

A more sophisticated version: other orderings / choose the next variable to update according to which one we expect to allow us to make the largest increase in $W(\alpha)$

9.2 SMO

Constraint: $\sum_{i=1}^m \alpha_i y^{(i)} = 0$

for $j \quad \forall i \neq j$ hold $y^{(i)}$ fixed $\Rightarrow y^{(j)}$ fixed

$\Rightarrow$ we must update at least two α 's simultaneously

Repeat till convergence {

1. Select some pair α_i and α_j to update next (using a heuristic that tries to pick the two that will allow us to make the biggest progress towards the global maximum).
2. Reoptimize $W(\alpha)$ with respect to α_i and α_j , while holding all the other α_k 's ($k \neq i, j$) fixed.

}

i.e. hold $\alpha_3, \alpha_4, \dots, \alpha_m$ fixed

reoptimize $W(\vec{\alpha})$ wrt α_1 & α_2 .

$$\alpha_1 y^{(1)} + \alpha_2 y^{(2)} = - \sum_{i=3}^m \alpha_i y^{(i)} \triangleq \tilde{c}$$

also $(y^{(1)}, y^{(2)})$ lies within $[0, c] \times [0, c]$

exists lower-bound L and upper bound H . $\Rightarrow y^{(0)} \in [L, H]$

$$\alpha_i = (\tilde{z} - \alpha_i y^{(i)}) / y^{(i)} \quad (|y^{(i)}|^2 = 1)$$

$W(\alpha_1, \alpha_2, \dots, \alpha_m) = W(\tilde{z} - \alpha_i y^{(i)} / y^{(i)}, \alpha_1, \dots, \alpha_m)$ — some quadratic function in α_i

$$\Rightarrow \alpha_i^{\text{new}} = \begin{cases} H & \text{if } \alpha_i^{\text{new, unclipped}} > H \\ \alpha_i^{\text{new, unclipped}} & \text{if } L \leq \alpha_i^{\text{new, unclipped}} \leq H \\ L & \text{if } \alpha_i^{\text{new, unclipped}} < L \end{cases}$$

$\alpha_i^{\text{new, unclipped}}$ removes the constraint of $[L, H]$